18.15 Rank the compounds in each of the following groups in order of decreasing acidity:

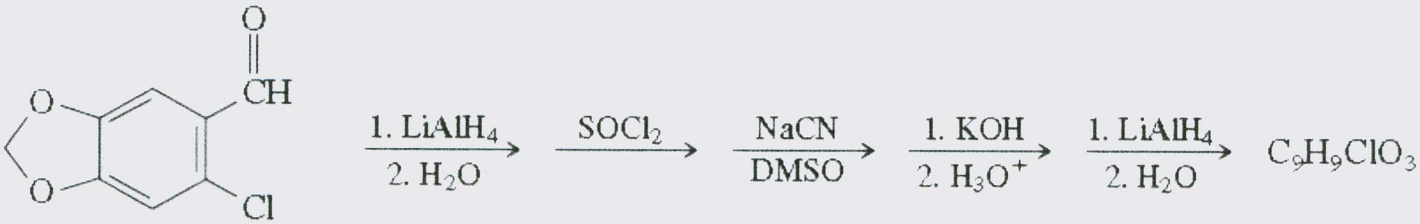
a) 1,3-Propanediol, propanedioic acid, propanoic acid

b) Acetic acid, ethanol, trifluoroacetic acid, 2,2,2-trifluoroethanol, trifluoromethanesulfonic acid (CF_3SO_2OH)

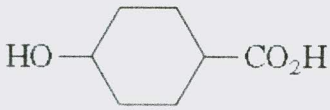
18.17 Propose methods for preparing butanoic acid from each of the following:

(e) 2-Propanol

18.22 The compound shown was subjected to the following series of reactions to give a product having the molecular formula $C_9H_9ClO_3$. What is this product?



18.24 (a) Which stereoisomer of 4-hydroxycyclohexanecarboxylic acid (cis or trans) can form a lactone? What is the conformation of the cyclohexane ring in the starting hydroxy acid? In the lactone?



18.27 When levulinic acid ($CH_3\overset{O}{\overset{\|}{C}}CH_2CH_2CO_2H$) was hydrogenated at high pressure over a nickel catalyst at 220°C, a single product, $C_5H_8O_2$, was isolated in 94% yield. This compound lacks hydroxyl absorption in its IR spectrum and does not immediately liberate carbon dioxide on being shaken with sodium bicarbonate. What is a reasonable structure for the compound?

Problems are from Carey and Giuliano, "Organic Chemistry" 8th edition, chapter 18.